

Template for Sensors interfacing with Arduino

1. Infrared (IR) Sensor (3 Marks)
2. Passive Infrared (PIR) Sensor (3 Marks)
3. Ultrasonic Sensor (3 Marks)
4. Teamwork (3 Marks)
5. LMS Submission (3 Marks)

Evaluation Criteria:

1. Connection diagram.
2. Code for Arduino for each sensor.
3. Successful execution of all four sensor activities.

Performance—12 Marks

Submission—03 Marks

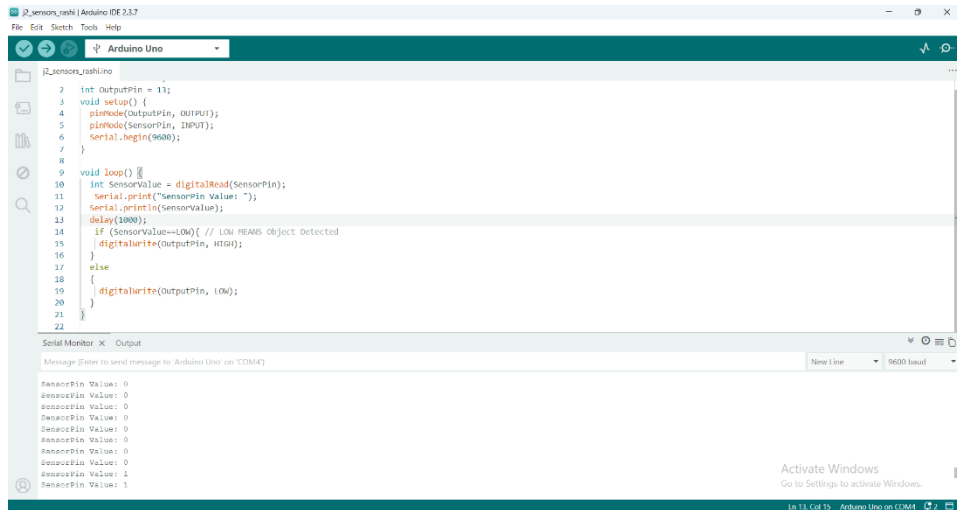
Team

Sr. No.	Roll No	Name	Work Done
1	16014325030	Avaneesh majumder	Connection
2	16014325031	Vijayta Malkani	Connection
3	160143250323	Sarbani mandal	Connection
4	16014325033	Om mane	Coding
5	16014325034	Aarav mody	Coding

Fill in your details as per the following points

- **Photos of Actual Implementation**
- **Video of Actual Implementation**
- **Conclusions**

1) IR SENSOR



```
1 int OutputPin = 11;
2
3 void setup() {
4   pinMode(OutputPin, OUTPUT);
5   pinMode(SensorPin, INPUT);
6   Serial.begin(9600);
7 }
8
9 void loop() {
10  int SensorValue = digitalRead(SensorPin);
11  Serial.print("SensorPin Value: ");
12  Serial.println(SensorValue);
13  delay(1000);
14  if (SensorValue==LOW) { // LOW MEANS Object Detected
15    digitalWrite(OutputPin, HIGH);
16  }
17  else
18  {
19    digitalWrite(OutputPin, LOW);
20  }
21 }
22
```

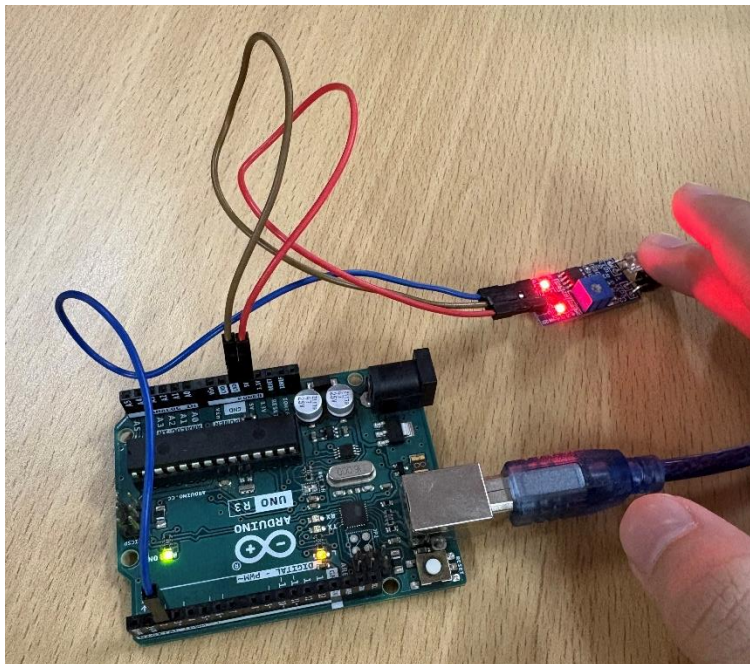
Serial Monitor x Output

Message (Enter to send message to 'Arduino Uno' on 'COM4')

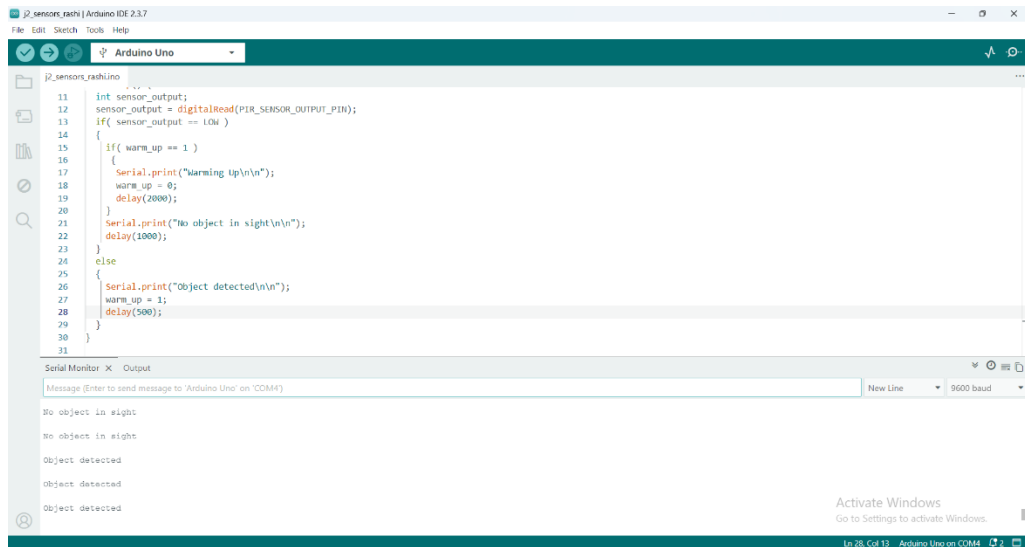
SensorPin Value: 0
SensorPin Value: 0
SensorPin Value: 0
SensorPin Value: 0
SensorPin Value: 0
SensorPin Value: 0
SensorPin Value: 0
SensorPin Value: 1
SensorPin Value: 1

Activate Windows
Go to Settings to activate Windows.

Ln 13, Col 15 Arduino Uno on COM4



2) PIR SENSOR

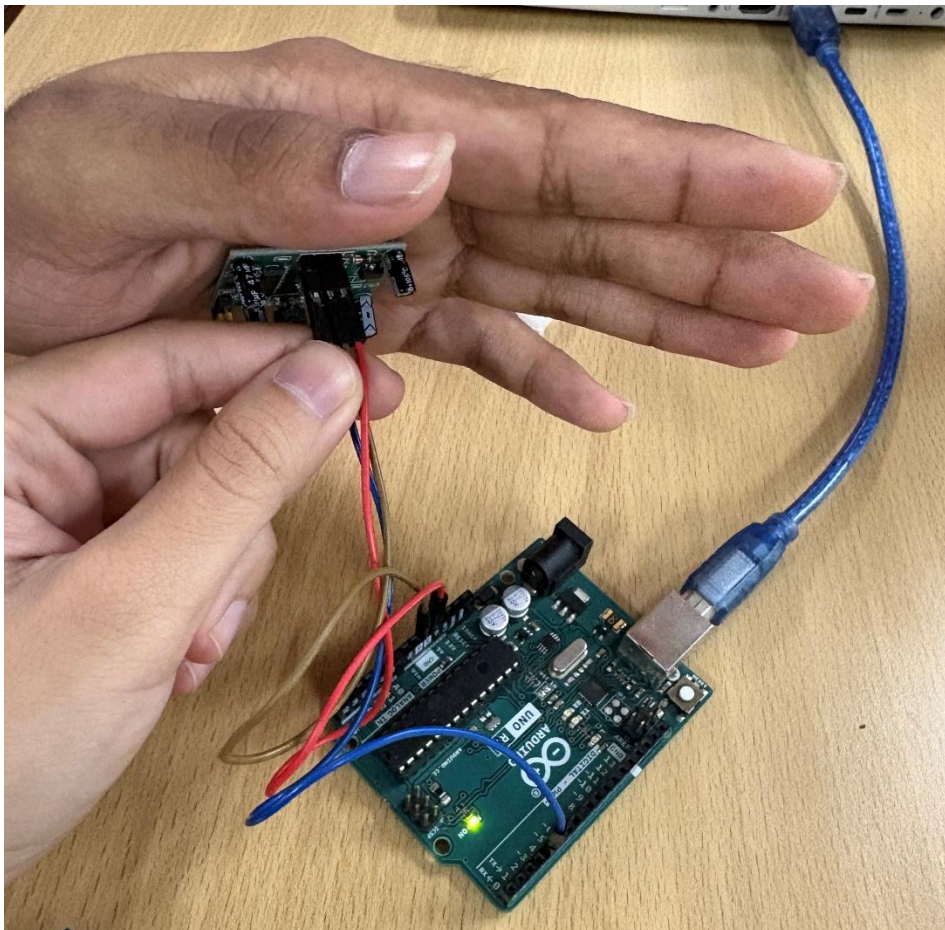


The screenshot shows the Arduino IDE 2.3.7 interface. The main window displays a C++ code for a PIR sensor. The code defines a variable `sensor_output` and uses `digitalRead` to read the PIR sensor's output. It includes a warm-up delay of 2000ms and a 1000ms delay between readings. The Serial Monitor at the bottom shows the output: "No object in sight", "Object detected", and "Object detected".

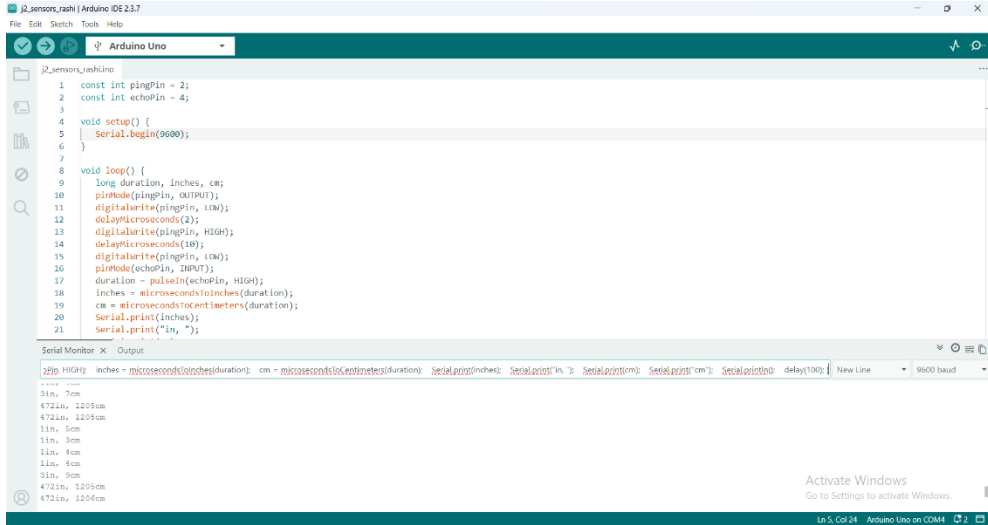
```
11 int sensor_output;
12 sensor_output = digitalRead(PIR_SENSOR_OUTPUT_PIN);
13 if( sensor_output == LOW )
14 {
15     if( warm_up == 1 )
16     {
17         Serial.print("Warning Up!\n");
18         warm_up = 0;
19         delay(2000);
20     }
21     Serial.print("No object in sight\n");
22     delay(1000);
23 }
24 else
25 {
26     Serial.print("Object detected\n");
27     warm_up = 1;
28     delay(500);
29 }
30 }
31 }
```

Serial Monitor Output:

```
Message (Enter to send message to 'Arduino Uno' on 'COM4')
New Line 9600 baud
No object in sight
No object in sight
Object detected
Object detected
Object detected
```



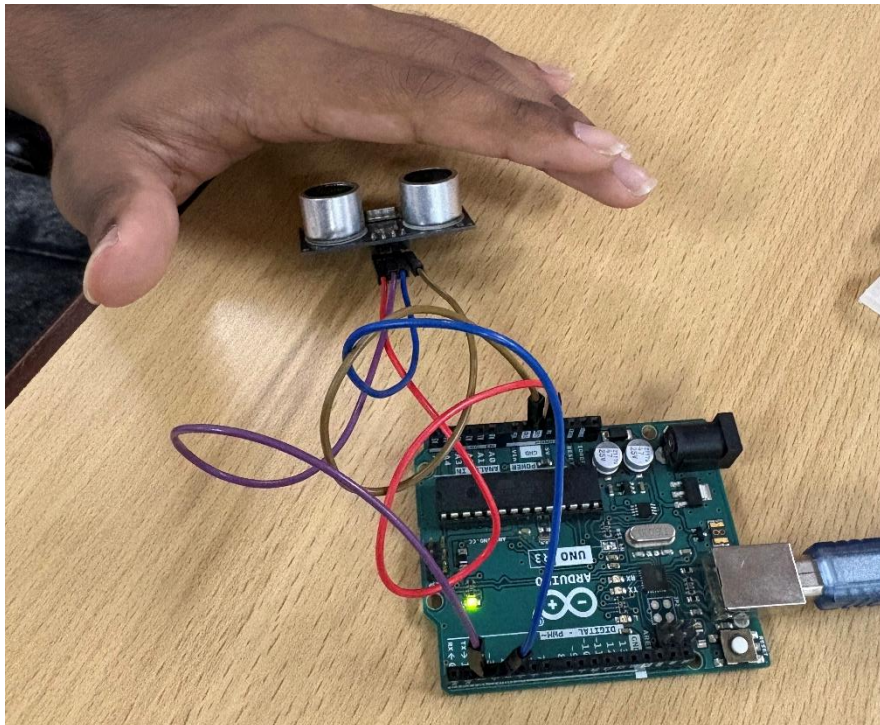
3) ULTRASONIC SENSOR



```
1 const int pingPin = 2;
2 const int echoPin = 4;
3
4 void setup() {
5   Serial.begin(9600);
6 }
7
8 void loop() {
9   long duration, inches, cm;
10  pinMode(pingPin, OUTPUT);
11  digitalWrite(pingPin, LOW);
12  delayMicroseconds(2);
13  digitalWrite(pingPin, HIGH);
14  delayMicroseconds(10);
15  digitalWrite(pingPin, LOW);
16  pinMode(echoPin, INPUT);
17  duration = pulseIn(echoPin, HIGH);
18  inches = microsecondsToInches(duration);
19  cm = microsecondsToCentimeters(duration);
20  Serial.print(inches);
21  Serial.print("in, ");
22  Serial.print(cm);
23  Serial.print("cm\n");
24  delay(1000);
25 }
```

Serial Monitor X Output

```
220in, HIGHinches = microsecondsToInches(duration) cm = microsecondsToCentimeters(duration) Serial.print(inches) Serial.print("in, ") Serial.print(cm) Serial.print("cm\n") delay(1000) New Line 9600 baud
3in, 7cm
472in, 1205cm
472in, 1205cm
1in, 5cm
1in, 3cm
1in, 4cm
1in, 4cm
3in, 9cm
472in, 1205cm
472in, 1205cm
```



GOOGLE DRIVE LINK FOR VIDEOS

https://drive.google.com/drive/folders/1fJv9fhOMh9ih9SJgrucQC4WkTHw7OqnB?usp=drive_link

CONCLUSION

In this experiment, we successfully tested and analyzed the working principles of IR, PIR, and Ultrasonic sensors. The IR sensor was effective in detecting nearby objects based on infrared reflection, making it suitable for short-range obstacle detection. The PIR sensor accurately detected motion by sensing changes in infrared radiation emitted by living beings, proving useful for security and motion-activated systems. The Ultrasonic sensor measured distance precisely making it ideal for distance measurement applications.